CSE445 Lecture 4.2

Principal Component Analysis (PCA)

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Important Concepts

Dimensionality Reduction, CoVariance Matrix, Eigenvectors, Eigenvalues

1. What Problem Does PCA Solve?

Many datasets contain many features. Some features may be correlated, noisy, or redundant.

PCA: Many correlated variables $\Rightarrow$ fewer informative directions

Principal Component Analysis (PCA) is an unsupervised dimensionality reduction method. It creates a smaller set of new variables called principal components.

The goal is simple: keep fewer dimensions while preserving as much variance as possible.

2. PCA Does Not Select Features

Feature selection keeps some original columns and removes others.

$$\text{Feature selection: } [x_1, x_2, x_3, x_4, x_5] \Rightarrow [x_2, x_5]$$

PCA is feature extraction. It creates new features by mixing the original features.

$$\text{PC}_1 = w_1 x_1 + w_2 x_2 + \cdots + w_d x_d$$

So PCA does not ask, "Which original columns should we keep?" It asks, "Which new directions summarize the data best?"

A weight can be close to zero, so one feature may contribute little. However, PCA still transforms the feature space instead of selecting columns in the usual sense.

3. Data Matrix and Standardization

Assume there are m samples and d features.

$$X \in \mathbb{R}^{m \times d}, \qquad X_{ij} = \text{value of feature } j \text{ for sample } i$$

Before applying PCA, we usually standardize each numeric feature:

$$Z_{ij} = \frac{X_{ij} - \mu_j}{\sigma_j}$$

Here μ_j is the mean of feature j, and σ_j is its standard deviation.

This step is important when features have different units or scales. Without standardization, a large-scale feature may dominate the principal components.

4. Covariance: How Features Move Together

After standardization, PCA studies how features vary together using the covariance matrix.

$$C = \frac{1}{m-1} Z^T Z, \qquad C \in \mathbb{R}^{d\times d}$$

Covariance sign	**Meaning**
Positive	Two features tend to increase or decrease together
Negative	One feature tends to increase when the other decreases
Near zero	Weak linear relationship between the two features

The covariance matrix tells PCA where the data has strong shared movement. PCA then finds directions that capture the largest spread of the data.

5. Eigenvectors Become Principal Component Directions

PCA finds eigenvectors and eigenvalues of the covariance matrix:

$$Cv_k = \lambda_k v_k$$

Here v_k is an eigenvector. It gives a principal component direction. Along this direction, the covariance matrix only stretches or shrinks the vector.

And, λ_k is an eigenvalue. It tells how much variance is captured along direction v_k.

The principal components are ordered by magnitude of the eigenvalues:

$$\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq \cdots$$

So PC1 captures the largest variance, PC2 captures the next largest variance, and so on. Moreover, PC2 is orthogonal to PC1, and similarly the other PCs.

6. Projecting Data onto Principal Components

After selecting the first r principal directions, form a matrix with them:

$$V_r = [v_1 \; v_2 \; \cdots \; v_r], \qquad V_r \in \mathbb{R}^{d \times r}$$

The new PCA features, also called PC scores, are obtained by projection:

$$Y = ZV_r, \qquad Y \in \mathbb{R}^{m \times r}$$

Each row of Y is the same sample represented using fewer dimensions. Each column of Y is a principal component feature, such as PC1 or PC2.

Dimension Reduction

$$X \in \mathbb{R}^{m \times d} \Rightarrow Y \in \mathbb{R}^{m \times r}, r \lll d$$

This is why PCA is useful for visualization, compression, and noise reduction.

7. Calculating PCA Using SVD

The same low-rank notation used in matrix factorization using SVD can describe PCA. For the standardized data matrix Z, SVD gives:

$$Z \approx U_r \Sigma_r V_r^T$$

$$\underbrace{Z}_{\text{standardized data}} \approx \underbrace{U_r}_{\text{sample positions}} \underbrace{\Sigma_r}_{\text{component strength}} \underbrace{V_r^T}_{\text{feature directions}}$$

Part	Meaning in PCA
U_r	Coordinates of samples in the reduced PCA space
Σ_r	Strength of the selected principal components
V_r^T	Principal directions written in terms of the original features

8. Explained Variance and Choosing r

PCA does not automatically decide how many components are enough. We choose r based on how much variance we want to keep.

$$\text{Explained variance ratio of } \mathrm{PC}_k = \frac{\lambda_k}{\lambda_1 + \lambda_2 + \cdots + \lambda_d}$$

$$\text{Cumulative explained variance for first } r \text{ PCs} = \frac{\lambda_1 + \lambda_2 + \cdots + \lambda_r}{\lambda_1 + \lambda_2 + \cdots + \lambda_d}$$

A common practical rule is to keep enough components to preserve a high percentage of variance, such as 90% or 95%, depending on the task.

Final idea: PCA replaces many original features with fewer orthogonal directions that keep most of the data's important variation.